

VAE Cheat Sheet

1. Denoising Autoencoders (DAEs) & Score Matching

Equation 1.1: The Forward Corruption

$$\tilde{x} = x_0 + \epsilon$$

- **What it does:** Adds Gaussian noise to a clean image to push it off the data manifold.
- **Exam Keywords:** "corrupt the input," "apply noise," "given a noisy observation."
- **Intuition:** ϵ is usually sampled from $\mathcal{N}(0, \sigma^2 I)$.

Equation 1.2: DAE Reconstruction Loss (MSE)

$$Loss = \frac{1}{N} \sum_{i=1}^N (\tilde{x}_i - x_i)^2$$

- **What it does:** Calculates the penalty for failing to predict the original clean image.
- **Exam Keywords:** "calculate the empirical risk," "find the L2 loss," "compute the Mean Squared Error for the denoising task."

Equation 1.3: Tweedie's Formula (Score Matching Bridge)

$$E[x|\tilde{x}] = \tilde{x} + \sigma^2 \nabla \log P(\tilde{x})$$

- **What it does:** Uses the score vector (the gradient of the log-density) to map a noisy image back to the clean data manifold .
 - **Exam Keywords:** "calculate the expected value of the clean data," "find the posterior mean," "use the learned score function to denoise," "apply Tweedie's formula."
 - **Intuition:** $\tilde{x}$ is where you are. $\nabla \log P(\tilde{x})$ is the direction of the clean images. σ^2 is how big of a step you take.
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2. Variational Autoencoders (VAEs)

Equation 2.1: The Reparameterization Trick

$$z = \mu_\phi(x) + \sigma_\phi(x) \odot \epsilon$$

- **What it does:** Decouples the random sampling ($\epsilon \sim \mathcal{N}(0, I)$) from the network's deterministic parameters (μ_ϕ, σ_ϕ) so backpropagation can occur .

- **Exam Keywords:** "calculate the latent vector z passed to the decoder," "perform a forward pass through the sampling layer," "evaluate the reparameterized state."

Equation 2.2: The Master ELBO Objective (Negative Form)

$$-L_{ELBO}(\phi, \theta) = -E_{q_\phi}[\log P_\theta(x|z)] + D_{KL}(q_\phi(z|x) || P_\theta(z))$$

- **What it does:** The total loss function of the VAE. We minimize the negative ELBO to maximize the true data likelihood.
- **Exam Keywords:** "calculate the total loss," "evaluate the Evidence Lower Bound," "compute the objective function."

Equation 2.3: Analytical KL Divergence (For 1D / Diagonal Gaussians)

$$D_{KL} = -\frac{1}{2} \left(1 + \log(\sigma_\phi^2) - \mu_\phi^2 - \sigma_\phi^2 \right)$$

- **What it does:** Computes the exact penalty for how far the encoder's output $\mathcal{N}(\mu_\phi, \sigma_\phi^2)$ is from the prior $\mathcal{N}(0, 1)$.
- **Exam Keywords:** "compute the regularization term," "find the KL divergence penalty," "evaluate the prior matching loss."
- **Intuition:** If $\mu_\phi = 0$ and $\sigma_\phi^2 = 1$, the math perfectly cancels out to 0.

Equation 2.4: Gaussian Negative Log-Likelihood (The "Blurriness" Equation)

$$-\log P_\theta(x_i|z) = \frac{1}{2} \left[\frac{(x_i - \mu_i)^2}{\sigma_i^2} + \log(\sigma_i^2) \right]$$

- **What it does:** Calculates the reconstruction loss assuming the decoder outputs a Gaussian distribution.
- **Exam Keywords:** "calculate the NLL reconstruction loss," "prove why the output is blurry," "evaluate the Gaussian decoder penalty."
- **Intuition:** If the variance σ_i^2 is forced to be large by the optimizer to minimize the fraction, the image becomes blurry.

Equation 2.5: The Posterior Collapse Condition

$$q_\phi(z|x) = P_\theta(z) \implies D_{KL} = 0$$

- **What it does:** The mathematical definition of a collapsed encoder .
- **Exam Keywords:** "diagnose the state of the encoder," "prove posterior collapse has occurred," "why is the latent space uninformative?"

3. Hierarchical VAEs (HVAEs)

Equation 3.1: HVAE Markovian Factorization (Inference & Generation)

Inference (Encoder): $q_\phi(z_1, z_2, z_3|x) = q_\phi(z_1|x)q_\phi(z_2|z_1)q_\phi(z_3|z_2)$

Generation (Decoder): $P_\theta(x, z_1, z_2, z_3) = P(z_3)P_\theta(z_2|z_3)P_\theta(z_1|z_2)P_\theta(x|z_1)$

- **What it does:** Breaks down the joint probability into step-by-step conditional probabilities.
- **Exam Keywords:** "write the joint distribution," "factorize the inference model," "apply the Markovian assumption."

Equation 3.2: The Chained ELBO

$$-L_{ELBO} = -E_{q_\phi}[\log P_\theta(x|z_1)] + E_{q_\phi}[D_{KL}(q_\phi(z_1|x)||P_\theta(z_1|z_2))] + E_{q_\phi}[D_{KL}(q_\phi(z_2|z_1)||P_\theta(z_2|z_3))] + D_{KL}(q_\phi(z_3|z_2)||P_\theta(z_3))$$

- **What it does:** The full loss function for an $L = 3$ HVAE, aligning every step of the encoder to every step of the decoder .
- **Exam Keywords:** "write the ELBO for the hierarchical model," "calculate the total loss across all latent layers."

4. High-Probability Exam Combinations

Professors rarely test one equation in isolation. Here are the expected combinations you will see in a 3-hour open-book setting:

Combo A: The "Full Forward Pass" (Eq 2.1 → Eq 2.4 → Eq 2.3 → Eq 2.2)

- **The Scenario:** You are given an input x_0 , the encoder parameters (μ_ϕ, σ_ϕ) , sampled noise ϵ , and the decoder parameters (μ_i, σ_i) .
- **The Flow:**
 1. Use the Reparameterization Trick (2.1) to find z .
 2. Use the Gaussian NLL (2.4) to find the reconstruction loss based on the decoder's output.
 3. Use the Analytical KL Divergence (2.3) to find the penalty.
 4. Add them together using the Master ELBO (2.2) to find the total loss for that single forward pass.

Combo B: The "Score vs. MSE" Comparison (Eq 1.1 → Eq 1.2 → Eq 1.3)

- **The Scenario:** You are comparing a standard DAE to a Score-Matching approach.
- **The Flow:**
 1. Corrupt the input (1.1).
 2. Calculate the standard MSE loss if the network predicts the image directly (1.2).
 3. Use Tweedie's formula (1.3) to calculate the expected clean image if the network instead predicted the score vector. Compare the two outputs.

Combo C: The "Prove It's Broken" Scenario (Eq 2.3 → Eq 2.5 or Eq 2.4)

- **The Scenario:** You are given network outputs and asked to explain a visual defect in the generated images.
- **The Flow (Posterior Collapse):** You calculate the KL Divergence (2.3). You find it is exactly 0. You cite the Posterior Collapse Condition (2.5) to prove the autoregressive decoder has ignored the latent space.
- **The Flow (Blurriness):** You calculate the NLL (2.4) using a sharp variance and a loose variance, proving mathematically that the optimizer favors the loose variance to minimize the loss penalty, resulting in blur.